

Digital Asset Cryptography

Part 1 - Asset-Specific Threat Signals: ECC Attack Demonstrations & Resource Estimates

Compiled by the Spaced Research Ledger

Last updated: 2026-08-31

Nobody has broken a real cryptographic key with a quantum computer. Everything in this part sits against that fact. What exists instead is a contested prize demonstration and two rival estimates of the hardware a real attack would need. Alongside them sits a piece of arithmetic about how fast such an attack would have to run.

Bitcoin and Ethereum sign transactions with elliptic curve cryptography, ECC for short. A quantum computer running Shor's algorithm, a quantum method for finding private keys, could in principle derive a private key from a public one. Classical computers cannot do that at these key sizes. The open question is scale: how many qubits, of what quality, running for how long. The field's two current answers disagree by a factor of about fifty.

Section 1 covers the Q-Day Prize demonstration and the rebuttal that followed it. Section 2 covers Google Quantum AI's estimate of the hardware needed to break a Bitcoin key. Section 3 covers a far lower neutral-atom estimate and the dispute over it. Section 4 covers the arithmetic behind the widely quoted nine-minute figure.

1. The Q-Day Prize and Its Rebuttal

A security company paid a one-bitcoin bounty for the largest public quantum attack on ECC to date. Bitcoin developers then showed the quantum computer contributed nothing to the result.

Current State

The award is real and the dispute is about what it demonstrated. Project Eleven awarded its Q-Day Prize, a one-bitcoin bounty worth about \$78,000 at the time, to Giancarlo Lelli on 24 April 2026.[1] Lelli derived a 15-bit elliptic curve private key from its public key on a publicly accessible IBM quantum computer, using a variant of Shor's algorithm.[1] Project Eleven presented the result as a 512 times jump over the previous public demonstration.[2] The company called it the largest public demonstration to date of the attack class threatening Bitcoin, Ethereum and over \$2.5 trillion in ECC-secured assets.[2]

The rebuttal came on the record. Jonas Schnelli, a former Bitcoin Core maintainer, reproduced the key recovery in Python using purely random bits, and found the quantum computer's outputs statistically indistinguishable from random coin flips.[3] Independent researcher Yuval Adam confirmed this by swapping the IBM backend for the random-number generator in Linux and getting identical results. Coinkite founder Rodolfo Novak argued that the private key is classically solved before the quantum circuit even runs, and Jimmy Song also criticised the result.[3]

The mechanism of the objection is simple. A 15-bit key lives in a search space of only 32,767 candidates.[1] That is small enough for classical checking against the public key to find a match by near-random sampling with high probability, so the quantum stage contributes no advantage.[3]

Project Eleven's chief executive Alex Pruden conceded the point in part. He acknowledged the result was not Q-Day, agreed that experiments on today's noisy hardware depend on classical assistance, and reframed the prize as incremental progress on accessible machines.[3] He maintained that planning for post-quantum migration remains prudent.[3]

2. Google's Resource Estimate

Google Quantum AI worked out how big a quantum computer would need to be to break the curve Bitcoin uses. The answer came in twenty times smaller than earlier estimates, and still far beyond any machine that exists.

Current State

The paper, "Securing Elliptic Curve Cryptocurrencies against Quantum Vulnerabilities", was announced on 30 March 2026 by researchers from Google Quantum AI, the Ethereum Foundation and Stanford University.[4] It gives two circuit designs for breaking a 256-bit key on secp256k1, the curve Bitcoin and Ethereum use. One needs at most 1,200 logical qubits and 90 million Toffoli gates, a basic quantum operation used here as a cost unit. The other needs at most 1,450 logical qubits and 70 million Toffoli gates.[5]

A logical qubit is an error-corrected unit built from many physical qubits. Mapped onto a superconducting architecture with a physical error rate of 10^{-3} , the design needs fewer than 500,000 physical qubits, roughly twenty times fewer than previous estimates.[5] No machine with even 1% of that capacity exists, and the paper is a resource estimate, not a timeline prediction.[5]

3. The Neutral-Atom Counter-Estimate

One day after Google's paper, a startup and two universities claimed the same attack could run on roughly ten thousand qubits. The claim is serious, self-interested, and unreviewed.

Current State

The paper "Shor's algorithm is possible with as few as 10,000 reconfigurable atomic qubits" appeared on 31 March 2026, from Oratomic, Caltech and UC Berkeley.[6] That was one day after the Google paper.[6]

Its figures depend on the architecture chosen. A space-efficient design needs 9,700 to 11,000 physical qubits, a balanced one 12,000 to 13,300, and a time-efficient one about 26,000.[6] Runtimes range from 10 to 264 days depending on parallelisation, assuming a cycle time of one millisecond.[6] The authors state that a machine with about 26,000 of these qubits could break a 256-bit ECC key in about ten days.[7]

The fifty-fold reduction against Google's figure comes from error correction. The authors use high-rate quantum LDPC codes, an error-correction family that spends far fewer physical qubits per logical qubit. Their

codes reach about 30% encoding efficiency, against about 4% for the surface codes in Google's mapping, and their lifted-product code encodes 1,480 logical qubits in 5,278 physical ones.[6]

The paper's own caveats are substantial. The authors describe their construction as a theoretically consistent existence proof that may not be optimal, and checking it became computationally prohibitive for the largest codes.[6] The assumed one-millisecond cycle time is undemonstrated, with current systems running several milliseconds per cycle on far fewer qubits, and individually addressed qubits at that scale lack experimental proof.[6] Some of the underlying codes were found by a search method assisted by large language models, which the community has yet to reproduce independently.[6]

Two further facts frame the claim. Current neutral-atom systems control 6,100 atoms, so the qubit-count gap is roughly a factor of two, far smaller than the gap facing superconducting approaches.[6] And all nine authors are shareholders in Oratomic, with six employed there, which makes the paper both a scientific result and a roadmap for the company's own hardware. It had not passed peer review as of the research behind this part.[7]

Unresolved Conflicts

The two papers disagree by a factor of about fifty on the hardware a Bitcoin key break requires. Google's mapping puts it under 500,000 physical qubits on superconducting hardware.[5] Oratomic and its academic co-authors put it near 10,000 reconfigurable atomic qubits, using much denser error correction.[6] Independent reproduction of the code constructions, and peer review of the neutral-atom paper, would settle whether the lower figure stands.[7]

4. The Exposure Window and the Odds

The widely quoted claim that a quantum computer could crack a Bitcoin key in nine minutes is real arithmetic resting on ideal assumptions. The odds of the required machine existing soon are polled, not known.

Current State

The nine-minute figure derives from the Google paper's circuit execution time. Assuming microsecond-scale gate operations and accounting for overheads, the full circuit runs on the order of tens of minutes. Because portions can be precomputed before a specific key is targeted, the target-specific phase comes to about nine minutes.[8]

Nine minutes matters because it sits just inside Bitcoin's average block time of roughly ten minutes.[8] An attacker that fast could in principle break a key from a public key revealed in the mempool, the waiting area for unconfirmed transactions, before the transaction confirms.[8] This on-spend attack is one modality among several, not the general case.[8] And the conditions are idealised: the figure depends on unproven architectures and assumes error-correction capability that does not exist.[8]

On probability, the Global Risk Institute's 2025 expert survey places such capability within ten years at quite possible, meaning 28% to 49%.[8] It only reaches likely, meaning 51% to 70%, at fifteen years.[8] The survey treats 2030 to 2035 as a reference window for migration readiness rather than a predicted arrival date.[8]

Researchers affiliated with the Ethereum Foundation, co-authors of the Google paper, estimated at least a 10% chance that a capable machine could exist by 2032.[9]

About This Report

The Spaced Research Ledger is an automated system that gathers claims from live public sources, checks each one against its origin, and records every disagreement instead of merging it. Every stage runs on a different AI model, drawn from three to seven systems across competing labs, so no single model both finds a claim and judges it. The Spaced Research Ledger is designed and maintained by Ridley DiSiena.

Sources

1. Project Eleven Awards 1 BTC Q-Day Prize (Project Eleven blog) - <https://blog.projecteleven.com/posts/project-eleven-awards-1-btc-q-day-prize-for-largest-quantum-attack-on-elliptic-curve-cryptography-to-date> (as of 2026-04-24)
2. blog.projecteleven.com — project eleven awards 1 btc q day prize for largest quantum attack on elliptic curve cryptography to date - <https://blog.projecteleven.com/posts/project-eleven-awards-1-btc-q-day-prize-for-largest-quantum-attack-on-elliptic-curve-cryptography-to-date> (as of 2026-04-24)
3. news.bitcoin.com — ibm quantum hardware cracks 15 bit ecc key but bitcoin devs say random bits match the result - <https://news.bitcoin.com/ibm-quantum-hardware-cracks-15-bit-ecc-key-but-bitcoin-devs-say-random-bits-match-the-result/> (undated)
4. Securing Elliptic Curve Cryptocurrencies against Quantum Vulnerabilities (arXiv:2603.28846) - <https://arxiv.org/abs/2603.28846> (as of 2026-03-30)
5. Securing Elliptic Curve Cryptocurrencies against Quantum Vulnerabilities (arXiv:2603.28846) - <https://arxiv.org/abs/2603.28846> (as of 2026-03-30)
6. Shor's algorithm is possible with as few as 10,000 reconfigurable atomic qubits (arXiv:2603.28627) - <https://arxiv.org/abs/2603.28627> (as of 2026-03-31)
7. coindesk.com — quantum computers could break crypto wallet encryption with just 10 000 qubits researchers say - <https://www.coindesk.com/markets/2026/03/31/quantum-computers-could-break-crypto-wallet-encryption-with-just-10-000-qubits-researchers-say> (as of 2026-03-31)
8. safeheron.com — google quantum whitepaper - <https://safeheron.com/blog/google-quantum-whitepaper/> (undated)
9. thequantuminsider.com — blockrock updates bitcoin etf with broadened warning about quantum computing - <https://thequantuminsider.com/2025/05/13/blockrock-updates-bitcoin-etf-with-broadened-warning-about-quantum-computing/> (undated)